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# Seroprevalence and trend of hepatitides among blood donors in a district hospital in Ghana: a nine-year retrospective, descriptive cross-sectional study

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**ABSTRACT**

Hepatitis infections are among the greatest threats to human existence and survival and hence poses a serious public health challenge in blood donation. This study aimed to determine the prevalence and trend of hepatitis infections among voluntary blood donors at Nkenkaasu District Hospital. A retrospective study of voluntary blood donors' records from January 2010 to December 2018 was conducted. Data on the donors' age, sex and serum HBsAg and Hepatitis C virus results were retrieved from the Laboratory's register and analyzed with SPSS version 16.0. Out of the 3306 total donors, the seroprevalence of hepatitis C infection was 11.7% and hepatitis B infection was 10.3%. Only 1.4% of the donors were co-infected with both pathogens and 82.7% had no infection. Rhesus blood group had a significant association with Hepatitis B ($P = 0.005$). As at 2010, both hepatitis infections had the same prevalence (73), from 2011 to 2018 both showed inconsistent trends. The relatively high seroprevalence of hepatitis infections identified from the study poses a greater threat to blood safety. Extensive screening of blood donors using standard techniques is highly recommended to ensure that recipients receive safe blood.

**KEYWORDS**

Hepatitides; blood donor; hepatitis B virus; hepatitis C virus

## Introduction

Blood transfusion is an essential life-saving measure in healthcare delivery as it offers supportive care for both medical and surgical patients.[1] Some population groups are at considerably higher risk of contracting infections associated with transfusion. These groups include children mostly under 5 years of age, suffering from severe malaria and anemia, pregnant women with bleeding disorders, patients of major trauma, hemophilic and thalassemic patients, and patients

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with other hemoglobin disorders.[2] Transfusion recipients are likely to encounter transfusion-associated problems such as hepatitis B virus (HBV) and hepatitis C virus (HCV) infections.[3] One of the most common ways of transmitting HCV and HBV is through exposure to infective blood.[4] It is approximated that 240 million people have chronic HBV and 130–150 million people are infected with HCV globally. Africa contribute to about 8% of all HCV infections worldwide.[5,6] More than 780,000 people die every year from HBV infection and approximately 500,000 people die from HCV-related liver diseases.[1,6,7] A study conducted in Nigeria reported that, 12.5% of patients who receive blood transfusion are likely to be infected with hepatitis B virus.[8] Alomatu in 2016 asserted that the prevalence of HBV and HCV among blood donors in Koforidua, Ghana are 13.2% and 8.0%, respectively.[6] These pathogens can persist in the blood of the individual as a carrier state or can cause asymptomatic infection. Blood banks in the various laboratories perform screening procedures to prevent the transmission of these pathogens to recipients.[9] Ghana as a country has adopted a measure to ensure that blood transfused to patients are safe by screening donors for the four transfusion-transmissible diseases; HIV, HBV, HCV, and syphilis.[10] However, the donor-screening process is often poorly and incompletely performed due to the country's resource constraints.[4] The few previous studies that reviewed hepatitis pathogens among blood donors did not consider the trend of the infections over the years.[1,11–13] In addition, there is scanty data on the burden of the blood-borne pathogens among blood donors in the middle belt of the country. In the light of the above, we sought to establish the prevalence and the trend of hepatitis infections among voluntary blood donors in the Nkenkaasu District Hospital.

## Subjects and methods

### *Study design and setting*

This institution-based descriptive cross-sectional study analyzed retrospective donors' data from January 2010 to December 2018 at the Blood Bank of the Nkenkaasu District Hospital located in Offinso-North district in the Ashanti-region, Ghana with a total population of about 68,543. Nkenkaasu Government Hospital, a 100-bed capacity facility, is the main referral facility in the district and its neighboring towns and villages. This hospital provides Out and In-patients' services, internal medicine, surgery, obstetric, laboratory, pharmaceutical, public health, and reproductive health service. The hospital has male and female surgical and in-patient wards, labour ward, pediatrics ward, and emergency ward. The total land area of the district is about 945.9 square kilometers and lies between latitude: 7°20N. 6°50S" and longitude: 1°60W. 1°45E". Majority of the inhabitants of this district are farmers.[14]

### Study participants

Data from 3306 voluntary blood donors, from nine consecutive years (January 2010 to December 2018), were retrieved. Apparently, healthy individuals aged 17–65 years with weight >50kg, and having responded appropriately to a panel of pre-donation questions comprising a medical history, were eligible for the donation. Participants responded to questions intending to exclude pregnant women, individuals having experienced high-risk sexual behavior within 2 weeks prior to the intended donation, recipients of previous transfusion and people who have experienced jaundice or shown signs of hepatitis.

### Data collection

Data on voluntary blood donors' age, sex and serological results of HCV and HBV were retrieved from the Laboratory's Blood Bank register. Information obtained from the records section included age, marital status, sex, blood group, rhesus factor, and results of serological markers for hepatitis B and hepatitis C. Donors with incomplete records were excluded from the study.

### Laboratory testing

Hepatitis B surface antigen (HBsAg) and HCV were detected by rapid diagnostic test (DiaSpot®, USA). DIASpot HBsAg is a one-step immunoassay-based test for qualitative detection of hepatitis B surface antigen (HBsAg) in serum. This test has a relative sensitivity and specificity of 99 and 97.0%, respectively. DIASpot HCV-Ab test strips have a sensitivity of almost 99.9% and a specificity of 98.6%. The center performs predonation screening on all apparently healthy individuals and only those without any of the infectious pathogens are required to donate their blood. The algorithm used in the blood bank is assessed in Figure 1.

### Statistical analysis

Data obtained were entered in Microsoft excel and analyzed with Statistical Package for the Social Sciences (SPSS), version 16.0. Means, frequencies, and percentages were computed. Differences in proportions among the different subgroups and other categorical variables were compared using the Chi-square test or Fisher's exact test, where appropriate. The results were presented using tables, charts, and graphs. Cochran-Armitage trend test assessed hepatitis B and C prevalence time-trend from 2010 to 2018. A $p$ value of $< 0.05$ were considered statistically significant.

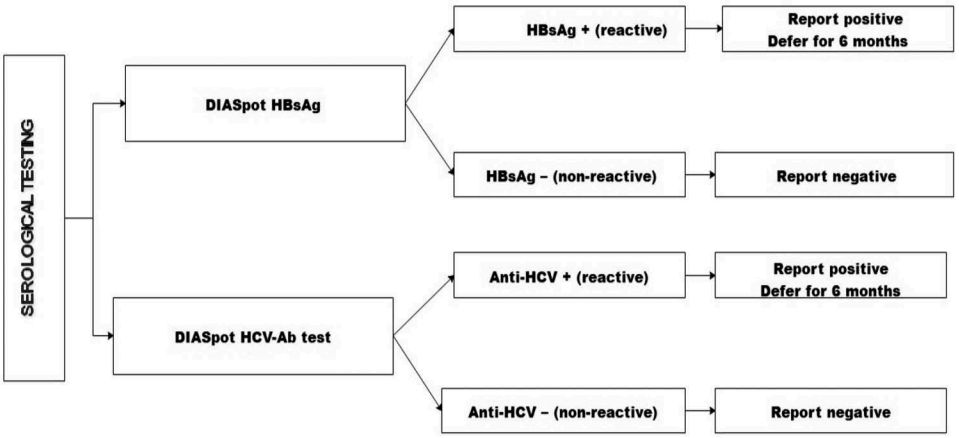



**Figure 1.** Summary of algorithm for serological screening for donor blood at Nkenkaasu District Hospital.

## Results

### *Demographic characteristics of the voluntary blood donors*

Table 1 shows demographic characteristics of study participants. Out of the voluntary blood donors, 2739/3306 (82.8%) were males and females were 567/3306 (17.2%) with female to male ratio of 1:4.8. The majority (72.5%) of the donors were below 36 years. The minimum and maximum ages of the donors were 17 and 65 years, respectively. More than half 2194/3306 (66.4%) of the donors were O ± . Almost all the blood donors were rhesus positive (98.8%) (Table 1).

### *Prevalence of hepatitis b and c infections with gender and age*

This study revealed that 332/2739 (12.1%) and 285/2739 (10.4%) of all males in this study were positive for Hepatitis C and B respectively. Also, 55/567 (9.7%) of the

**Table 1.** Demographic data of voluntary blood donors at Nkenkaasu District Hospital, January 2010 to December 2018.

| Variable | Number (n) | Percentage (%) |
|---|---|---|
| **Age** | | |
| <36 | 2397 | 72.5 |
| 36–45 | 711 | 21.5 |
| 46–55 | 180 | 5.4 |
| >55 | 18 | 0.5 |
| **Gender** | | |
| Male | 2739 | 82.8 |
| Female | 567 | 17.2 |
| **ABO blood group** | | |
| A+/– | 408 | 12.3 |
| B+/– | 674 | 20.4 |
| AB+/– | 30 | 0.9 |
| O+/– | 2194 | 66.4 |
| **Rhesus blood group** | | |
| Positive | 3100 | 98.8 |
| Negative | 206 | 6.2 |

female donors tested positive for HCV and 57/567 (10.1%) tested positive for HBV. Gender had no association with hepatitides prevalence ($p > .05$) (Table 2). About 11.4% and 11% of all donors aged < 36 years tested positive for HCV (273/239) and HBsAg (264/2397), respectively. Among individuals within 36–45 years age group 12.8% were reactive to HCV and 9.1% were reactive HBsAg. Also, some of the participants aged between 46 and 55 had positive results for HCV 21/180 (11.7%) and HBsAg 11/180 (6.1%). Only two persons of participants >55 years of age tested positive for HCV 2/180 (11.1%) and HBsAg 2/180 (11.1%). Age had no significant association with hepatitis C and B infection prevalence ($P > 0.05$) (Table 2).

### Distribution of hepatitis b and c infections with blood group

Regarding the distribution of the infections with blood group, 92/674 (13.6%) of donors with blood group B were positive for HCV, whereas 48/408 (11.8%) of HBsAg reactive donors belonged to blood group A. AB donors recorded the least prevalence of both HCV 3/30 (10.0%) and HBsAg 1/30 (3.3%). Rhesus positive (RH+) donors accounted for 11.8% of HCV infection and 10.4% for HBV infection. Rhesus negative RH− donors showed relatively low prevalence of HCV, 22/206 (10.7%), and HBsAg, 22/206 (10.2%). Rhesus had significant association with hepatitis B infection prevalence ($P = 0.005$) (Table 3). Maximum seropositive cases of HBV (45/383) were found among donors with blood group A positive, while the seroprevalence rates for HCV was highest in blood group B (83/632) positive (Table 4).

### Prevalence of hepatitis B and C coinfection among the blood donors

Prevalence of HCV and HBsAg infections were 387/3306 (11.7%) and 342/3306 (10.3%), respectively (Figure 2). Out of the 683 (20.7%) donors who had tested positive for the infections, 637/3306 (19.3%) were positive for only one pathogen.

**Table 2.** Distribution of hepatitides by sex and age among the voluntary blood donors at Nkenkaasu District Hospital, January 2010 to December 2018.

| | HCV | | HBsAg | |
|---|---|---|---|---|
| Variables | R N (%) | NR N (%) | R N (%) | NR N (%) |
| **Gender** | $P = 0.103$ | | $P = 0.802$ | |
| Male | 332 (12.1) | 2407 (87.9) | 285 (10.4) | 2454 (89.6) |
| Female | 55 (9.7) | 512(90.3) | 57(10.1) | 510 (89.9) |
| **Age (years)** | $P = 0$ .707 | | $P = 0$ .124 | |
| <36 | 273(11.4) | 2124 (88.6) | 264 (11.0) | 2133 (89.0) |
| 36–45 | 91 (12.8) | 620 (87.2) | 65(9.1) | 646 (90.9) |
| 46–55 | 21 (11.7) | 159(88.3) | 11(6.1) | 169 (93.9) |
| >55 | 2(11.1) | 16(88.9) | 2 (11.1) | 16 (88.9) |

Data are presented as frequencies with corresponding proportions in parentheses. HCV-Hepatitis C Virus and HBsAg-Hepatitis B Surface Antigen Virus. R-Reactive test, NR-Non-reactive test. *P* denotes *P*-value, * denotes significant *P*-value ($P < 0.05$).

**Table 3.** Distribution of hepatitides by blood type among donors at Nkenkaasu District Hospital, January 2010 to December 2018.

| | HCV | | HBsAg | |
|---|---|---|---|---|
| | R | NR | R | NR |
| Blood type | N (%) | N (%) | N (%) | N (%) |
| **ABO blood group** | $P = 0.365$ | | $P = 0.477$ | |
| A+/− | 47(11.5) | 361(88.5) | 48(11.8) | 360(88.2) |
| B+/− | 92 (13.6) | 582(86.4) | 63 (9.3) | 611(90.7) |
| AB+/− | 3 (10.0) | 27(90.0) | 1 (3.3) | 29 (96.7) |
| O+/− | 245(11.2) | 1949(88.8) | 230(10.5) | 1946(89.5) |
| **RH blood group** | $P = 0.126$ | | **$P = 0.005$*** | |
| Positive | 365 (11.8) | 2735 (88.2) | 321(10.4) | 2779 (89.6) |
| Negative | 22 (10.7) | 184 (89.3) | 21(10.2) | 185 (89.8) |

Data are presented as frequencies with corresponding proportions in parentheses. HCV-Hepatitis C Virus and HBsAg-Hepatitis B Surface Antigen, Virus. R-Reactive test, NR-Non-reactive test. *P* denotes *p*-value, * denotes significant *P*-value ($P < 0.05$).

**Table 4.** HBV and HCV reactivity in separate blood types among donors at Nkenkaasu District Hospital, January 2010 to December 2018.

| ABO/Rh Blood Group | HBV positive N (%) | HCV Positive N (%) | Total Subjects (%) |
|---|---|---|---|
| O positive | 215 (10.5) | 235 (11.4) | 450/2057 (0.23) |
| A positive | 45 (11.7) | 44 (11.5) | 89/383 (0.23) |
| B positive | 60 (9.5) | 83 (13.1) | 143/632 (0.23) |
| AB positive | 1 (3.6) | 3 (10.7) | 4/28 (0.14) |
| O negative | 15 (10.9) | 10 (7.3) | 25/137 (0.18) |
| A negative | 3 (10.9) | 3 (12.0) | 6/25 (0.24) |
| B negative | 3 (7.1) | 9 (21.4) | 12/42 (0.29) |
| AB negative | 0 (0.0) | 0 (0.0) | 0/2 (0.0) |

Data are presented as frequencies with corresponding proportions in parentheses. HCV-Hepatitis C Virus and HBsAg-Hepatitis B Surface Antigen

Exactly 46/3306 (1.4%) were co-infected with both hepatitis B and C and 2623 (79.3%) of the study participants were free from the two infections (Figure 3).

## *Trend of seroprevalence of HBV and HCV among the voluntary donors over the 9 years*

As at 2010, the prevalence of hepatitis B and C were at the same level, however, Hepatitis B declined from 2010 to 2011, whereas Hepatitis C declined steadily during that period. There was a general sharp decline in Hepatitis B and C from 2011 to 2013. Both hepatitis B and hepatitis C rose moderately to almost the same level between 2013 and 2014 after which hepatitis C showed a slight decline as against a sharp decline in hepatitis B from 2014 to 2015. There was a transient rise in hepatitis B from 2015 to 2016, but hepatitis C infection on the other hand, continued to decline till 2017. Both began to rise moderately from the year 2017 to 2018. Thus, hepatitides are currently declining as compared with 2010. Cochran-Armitage trend analysis showed both declines to be significant. The Cochran–Armitage trend test for hepatitis B infection prevalence resulted in

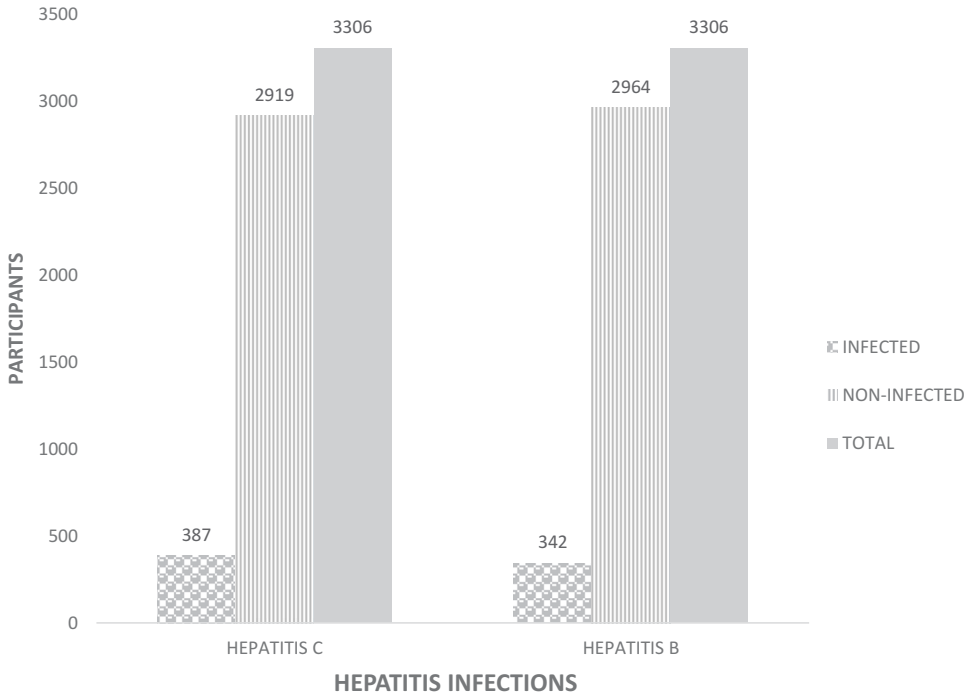



**Figure 2.** Prevalence of hepatitides among voluntary blood donors at Nkenkaasu District Hospital, January 2010 to December 2018.

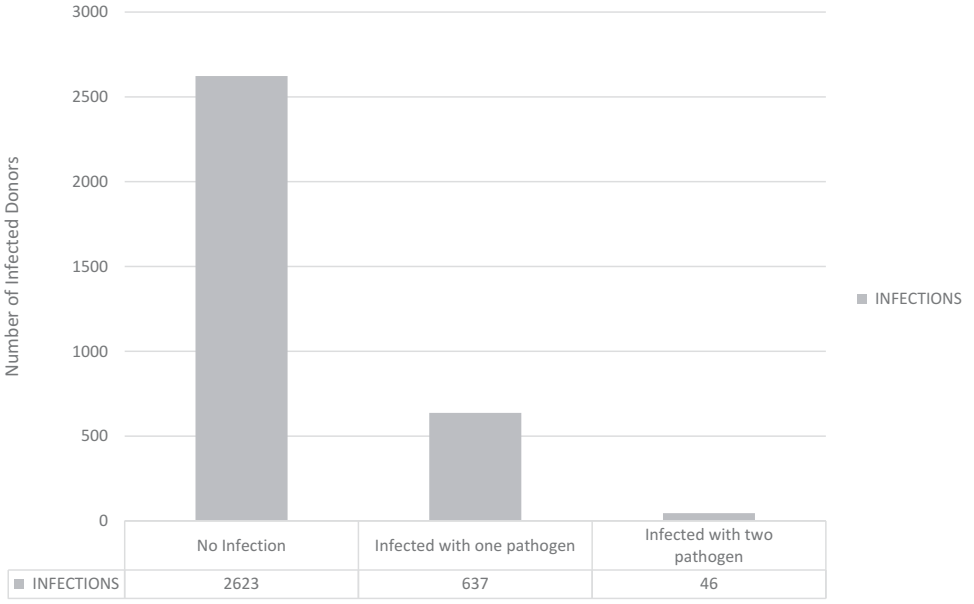



**Figure 3.** Infection status of blood donors by hepatitides pathogens at Nkenkaasu District Hospital, January 2010 to December 2018.

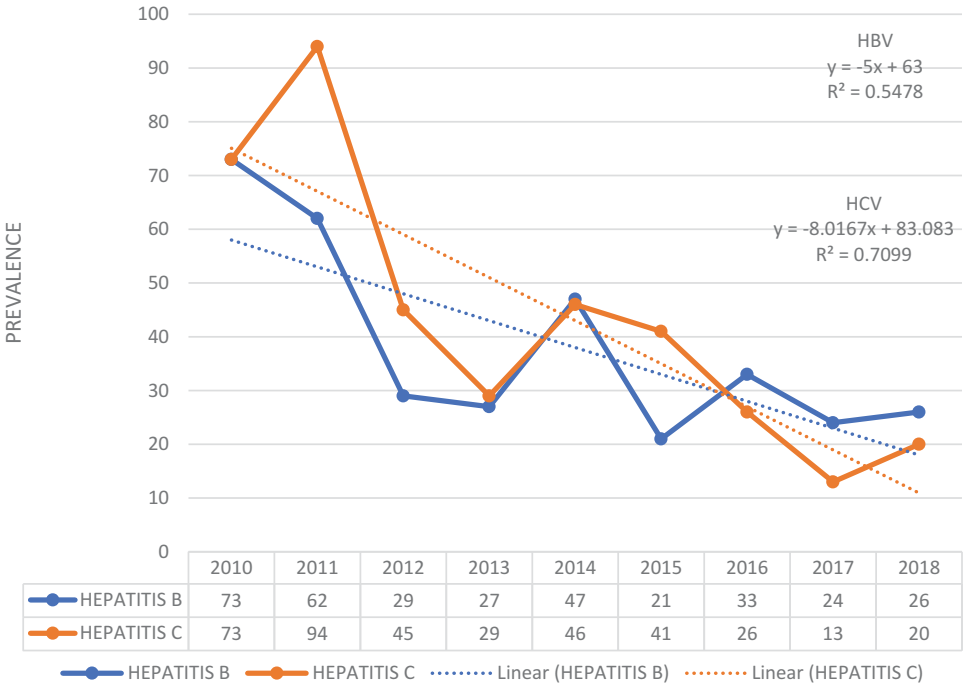



**Figure 4.** Shows the trend over the nine years of the hepatitides among the voluntary blood donors at Nkenkaasu District Hospital, January 2010 to December 2018. The Cochran–Armitage trend test for Hepatitis B infection prevalence resulted in −0.015 regression slope and a *P* value < 0.0001. Hepatitis C prevalence also showed a significant ($P < 0.0001$) decline according to the Cochran–Armitage test with a regression slope of −0.022.

−0.015 regression slope and a *P* value < 0.0001. Hepatitis C prevalence also showed a significant ($P < 0.0001$) decline according to the Cochran–Armitage test with a regression slope of −0.022 (Figure 4).

## Discussion

This retrospective, cross-sectional, descriptive study determined the prevalence and the trend of Hepatitis B and C among voluntary blood donors in Nkenkaasu District Hospital in Ashanti region, Ghana. In the present study, majority (82.8%) of the donors were males (Table 1). This is because more males qualify following the criteria of fitness and also higher proportion of males came for donations. Similar reasons can be ascribed to why majority (72.5%) of the blood donors were below 36 years of age. Most common blood group among the donors was O followed by B, the least common was AB as depicted in Table 1. Maximum seropositive cases of HBV (45/383) were found among donors with blood group A positive, while the seroprevalence rates for HCV was highest in blood group B (83/632) positive (Table 4). Rhesus had significant association with hepatitis B infection prevalence ($P = 0.005$) (Table 3). ABO and Rh groups' system is the

most important one among the 300 red cell antigens.[15] It is known the relationship between ABO blood groups and some infectious diseases such as malaria, *Helicobacter pylori* infections, cholera, and Norwalk virus infections.[15]The responsible pathogenesis for this susceptibility is that as many organisms that may bind to polysaccharide on cells and soluble blood group antigens may block this binding.[16] Additionally, some microbes share (molecular mimicry) blood group antigens with their hosts.[17] However, it is not clear if ABO/Rh blood groups have a role in the development of HBV and HCV infection. Many studies available do not have statistical data about the relationship between ABO/Rh blood groups and HBV and HCV infection.[18–22] In all these studies only the HBV and HCV infection rates in every blood group type were reported.

The seroprevalence of hepatitis C among the donors in this study is 11.7%. The seroprevalence of hepatitis C identified in our study is considerably high. Several previous studies conducted elsewhere in African sub-regions such as: 4.8% in Edéa, Cameroon[23], 1.1% in Nigeria[24], 0.51%[25] and 8% in Ghana[26] recorded much lower prevalence rates. The prevalence of hepatitis B among the blood donors in this study is 10.3%, which is higher than findings from elsewhere, 6.2% in Northwest Ethiopia[27], 5.3% in Nigeria[24], 5% in Sudan[28], 3.91% in Islamabad[29] and 0.63%.[30] The current finding is similar with a prevalence of 9.5 and 10.1%, reported in studies carried out in Southern Ethiopia and Cameroon, respectively.[3,23] On the contrary, the prevalence of hepatitis B in this study is far lower than a prevalence of 25% in North Ethiopia[2], 14.9% in Burkina Faso[31], and 14.3% in Jos, Nigeria.[32] This relatively high prevalence of HBV and HCV calls for stricter screening of donated blood and implementation of more stringent donor selection criteria. The higher prevalence of hepatitides recorded could be due to the fact that most participants were mostly replacement and commercial blood donors. Researches performed in several countries have proven that blood collected from volunteer, non-remunerated donors is much safer than blood collected from replacement donors.[33] Replacement donation accounts for 75 to 80% of blood for transfusion in Africa. HBV and HCV infections are most likely to be recorded in blood from replacement donors because relatives are most likely to deny certain risk lifestyles.[34] Ghana has a few centralized blood centers to support voluntary non-remunerated blood donation hence replacement donation is the main pattern of blood donation at the facility. The observed variations in the prevalence of both Hepatitis B and C could also be due to actual changes in population risks or effectiveness of donor screening measures. Most studies in sub Saharan Africa[15,33,35,36] do have prevalence of HBV greater than HCV but the result of this study showed otherwise. Although these pathogens share common modes of transmission and risk groups, the risk factors of the population under study directly influence the prevalence in that particular population.[33] Factors affecting the prevalence of HBV and HCV include behavioral, environmental and host factors, age at time of infection and mode of acquisition, which vary geographically[15,35]

Our study also determined prevalence of hepatitis B and C co-infections among the blood donors. The 19.3% of single pathogen infection identified among our study participants is lower relative to a study in Burkina Faso, which reported that the prevalence for single pathogen among blood donors was 29.82%[37] and 36.15% of one pathogen infection in a prospective study in Koforidua, Ghana.[6] The 1.4% co-infection rate observed in this study is comparable to studies by Nagalo in Burkina Faso and a prospective study in Ghana by Alomatu recording 3.30 and 2.35%, respectively.[6,31] However, our finding does not support a study by Farshadpour et al.[38] who concluded that co-infections of HBV and HCV among blood donors is normally of low prevalence in their 11-year retrospective study. Age, Sex, and ABO blood group had no significant association with Hepatitis B virus and HCV prevalence ($P > 0.05$). Similarly, a study conducted in Ethiopia found no associations between gender and hepatitis.[39] The prevalence of hepatitides fluctuated between 2010 and 2018 and time-trend analysis showed a significant decline over the years. this finding is consistent with a similar retrospective study at Gondar University Teaching Hospital, Northwest Ethiopia where HCV seroprevalence was observed declining among blood donors over a period.[28] The decline in hepatitis B and C infections over the years could partly be attributed to efforts made by Ghana Health Service to reduce this health menace, the focus has been on awareness creation, screening and availability of HBV vaccine. However, most Ghanaians have not adopted HBV and HCV preventive behaviors even though they are aware of the disease. Additionally, until recent years, health workers in the study site were not administering all the recommended pre-donation questionnaires to 'would be' blood donors and therefore whoever came there was screened for hepatitis, and this accounted for the inconsistent prevalence. However, the prevalence's started declining over the years when strict adherence to the pre-donation questionnaires to exclude potential risk donors was fully implemented in the facility. HBV and HCV were on same trend in 2010 but not thereafter, the possible explanation to this finding is the introduction of HBV vaccination at this facility around 2011. Also, the cost of the vaccine sometimes deters individuals from accessing the vaccines.[35] Although the vaccine remained unavailable and financially out-of-reach for many families in the peri- urban and rural areas, some individuals were able to afford it.

The seroprevalence of hepatitides identified from the blood donors was relatively high and this poses a great threat to blood safety. The various infections showed inconsistent trends over the years, however, they were seen declining with reference to 2010. Extensive screening of blood donors using standard techniques such as neutralization test, RIBA and western blot is highly recommended to ensure that recipients receive safe blood. However, the study was not without a limitation, the sequential pattern of screening of the donors' blood could underestimate the prevalence of co-infections among

them. Again, donors whose records were not complete in the Laboratory register were not included in the study.

## Authors' Contributions

This work was carried out in collaboration between all authors. Authors CN, SD, and FOB conceived and designed the study. Authors CN, FOB, and ROA collected the data. CN and SD analyzed, interpreted the data and drafted the manuscript. All authors critically reviewed, revised and approved the final manuscript.

## Competing Interests

All authors declare that there is no competing interest.

## Ethical Consideration

Permission for the study was obtained from the ethical committee of Nkenkaasu District Hospital. However, because of the retrospective nature of the study, informed consent could not be obtained from the study participants.

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